

Movement patterns of invertebrates in temporary and permanent streams

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Summary. Although it has been shown that invertebrates recolonize reflooded temporary streams from permanent refuges, e.g., the hyporheic zone, it has not been shown that they actively move into these refuges as streams dry. Substrate filled cages and drift nets were used to monitor invertebrate movement in two temporary streams and a permanent stream prior to and during drying to determine whether invertebrates leave drying riffles and enter flooded riffles. Invertebrate movement was essentially unidirectional in the permanent stream with downstream drift and within-substrate downstream movement dominating. In the temporary stream, movement vertically downward toward the hyporheic zone and upstream movement substantially contributed to a departure from a unidirectional pattern. In addition, prior to stream drying the relative colonization rate was higher and drift rate was lower in the temporary streams than in the permanent stream. During drying of the temporary stream, upstream movement continued to dominate but hyporheic movement was unimportant. Further, the upstream movement did not occur at the end of the riffle where it would lead to migration into non-drying riffles. Thus, even though movement patterns were different in permanent and temporary streams the pattern observed during stream drying would result in the concentration and subsequent death of invertebrates in drying riffles. This observation demonstrates that movement patterns of stream invertebrates do not necessarily result in behavioral avoidance of a dry period of temporary riffles.

Key words: Temporary streams – Stream invertebrates – Upstream movement – Hyporheic zone – Drift

As a temporary stream refloods it is colonized by invertebrates that come from the refuges inhabited during the dry period (Delucchi 1988a; McArthur and Barnes 1985; Williams 1977). Non-drying pools, the hyporheic zone and permanent streams are examples of such refuges. Alternatively, stream invertebrates may have survived the dry period as drought resistant forms in the substrate or as terrestrial adults that emerged from permanent streams or from the temporary stream before it dried. Williams (1977) compared four sources of colonization (drift, upstream movement, movement from the hyporheic zone and oviposition from

aerial adults) in two newly flooded temporary streams and a permanent stream and found that aerial colonizers were an unimportant source of colonists to temporary streams. Williams' (1977) study suggests, then, that migration to reflooded streams occurs primarily from adjacent refuges connected by water to the reflooded site.

Four general directions of invertebrate movement in relation to water flow have been studied (Williams 1981): (1) downstream, either by drift or within-substrate movement, (2) upstream, (3) perpendicular to flow, toward the banks or (4) down into the hyporheic zone (deep substrate). Drift is a common and often studied pattern of movement in streams (Waters 1972; Wiley and Kohler 1984). Studies testing whether upstream movement compensates for downstream drift have shown that even though upstream movement exists, numbers are low compared to the number of invertebrates drifting (6.5%, Bishop and Hynes 1969; 7–30% Elliott 1971). Further, drift is not the only source of downstream movement, since invertebrates may move downstream on or within the substrate. It has been suggested that positive rheotaxis in stream invertebrates would result in substantial upstream movement within or on the substrate (Bishop and Hynes 1969). However, comparisons of up-and-downstream within-substrate movement have shown that downstream movement is actually higher than upstream movement (Elliott 1971; Peckarsky 1979). Thus, upstream movement does not even compensate for within-substrate downstream movement. The few records of lateral movement in streams show movement of mature nymphs towards banks before emergence (Williams 1981).

Although the hyporheic zone is used extensively as a habitat by invertebrates (Williams 1984) little is known about the movement between the hyporheic zone and the surface of the streambed. Many have suggested that the deep interstices are a critical refuge for invertebrates during periods of unfavorable conditions on the surface of the substrate (Williams 1984); but few have actually documented this hypothesis experimentally. Sequential sampling of the hyporheos indicates that seasonal movements may occur, with mature insects migrating upward before emergence (Williams 1981). A recent study by Pugsley and Hynes (1986) showed that *Allocaenia pygmaea* moves freely between hyporheic zone and streambed, moving deeper and towards banks to diapause.

In addition to movement within riffles, invertebrates may migrate between riffles in one stream or from stream to stream. Although adult insects may emerge from a

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stream and fly (or walk) between riffles and streams, the migration of immature stages of insects and other invertebrates is limited to movement by the mechanisms listed above. The difference between within-riffle movement and migration is that migration involves movements over long distances and through different habitats, e.g., pools, that may not be suitable for invertebrates living in riffles. Although the immature forms of some insect species have been shown to make long upstream migrations into temporary streams each year (Neave 1930; Olsson and Soderström 1978), little else is known about migration of stream invertebrates. Work by Brussock (1986) indicates that if currents are above 10 cm/s invertebrates can drift through pools to adjacent downstream riffles; however, at lower currents they move upstream out of pools back into source riffles. Thus, a low flow regime may limit the extent of migration between stream riffles. Colonization of substrates placed in riffles has also been shown to be lowest during periods of lowest flow (Delucchi 1988a).

Although invertebrates colonize temporary streams through movement from refuges, it has not been determined whether invertebrates actively enter refuges as the stream dries or whether the streams are recolonized by invertebrates that happen to be in the refuges during stream drying. It would be advantageous for invertebrates in temporary streams to move actively into refuges (e.g., non-drying pools, permanent streams, hyporheic zone) before or as the stream dries. There is some evidence to suggest that invertebrates actively leave areas as they began to dry (Delucchi 1988a). Also, drift has been shown to increase as a response to drastic decreases in current (Minshall and Winger 1968; McArthur and Barnes 1985). However, not all studies have shown invertebrates actively leaving substrates placed in decreased current (Delucchi 1988a; Ulfstrand 1968).

The purpose of this study was to examine patterns of within-substrate movement and drift of invertebrates in temporary streams just prior to stream drying, to determine whether invertebrates leave riffles and enter flooded refuges as the stream dries. Experiments were designed to measure (1) within-riffle movement, (2) the extent of movement from riffles to the hyporheic zone, and (3) migration between riffles (movement from riffles into pools, non-drying riffles and permanent streams). Identical experiments were also carried out in a permanent stream to determine whether patterns found in the temporary streams were unique.

Methods

Study sites

Studies were conducted in three streams in a single watershed in the Ellis Hollow area of Tompkins County, New York. All streams have similar substrates of mainly gravel and cobble-sized stones, but with components ranging in size from silt to boulders. The three streams varied in permanence (Table 1). Cascadilla Creek is permanent along its entire length; Hunt Hill Creek is intermittent having both permanent riffles and riffles that dry; Hurd Road Creek is temporary, becoming dry along its entire length. Three types of riffles classified according to riffle permanence (Table 1) were used in the study.

Within-riffle movement experiment

In summer 1984, identical experiments were run at the five different riffle types including 2 permanent, 2 temporary and 1 dry riffle (Table 2). Five types of cages (Table 3) were used to test for invertebrate movement in four direc-

Table 1. Explanation of riffle and stream types used in movement experiments

Stream name	Discharge ^a (m ³ /s)	Stream type	Riffle types present	Definitions
Cascadilla Creek	0.0041–0.081	Permanent	Permanent	Flows all year
Hunt Hill Creek	0–0.019	Intermittent	Permanent Temporary Dry	Flows all year Flows more than 9 months Flows less than 9 months
Hurd Road Creek	0–0.0054	Temporary	Temporary	Flows more than 9 months

^a Range of summer discharge

Table 2. Dates and ranges of abiotic conditions measured at cages for the within-riffle movement experiment (summer 1984). Median change in depth and current over the study are given for periods where significant changes occurred (Wilcoxon signed rank test). N.S., not significant. Definitions for riffle and stream types are given in Table 1

Stream type	Permanent	Intermittent	Temporary		
Riffle type	Permanent	Permanent	Temporary	Dry	Temporary
Dates	1–4 August	9–12 July	7–10 July	20–23 June	22–25 June
Range of starting current (m/s)	0.12–0.52	0.08–0.73	0.11–0.8	0.1–0.88	0.05–0.38
Range of ending current (m/s)	0.14–0.56	0.06–0.78	0.12–0.9	0.07–0.36	0.05–0.36
Median change in current (m/s)	N.S.	N.S.	–0.045	–0.065	N.S.
Range of starting depths (cm)	5.0–19.0	4.5–22.0	4.0–18.0	5.0–13.0	1.5–6.0
Range of ending depths (cm)	4.5–19.0	4.0–20.0	3.0–16.0	3.0–9.5	1.5–5.5
Median change in depth (cm)	–0.25	–1.0	–2.0	–2.75	N.S.

Table 3. Description of cage types for the within-riffle movement experiment

Cage types and descriptions	no of sides with screen	Type of movement allowed
UD Up-and-downstream 2 sides open		Up-and-downstream movement within the substrate
TUD Top and up-and-downstream sides open	3	Drift and up-and-downstream movement within substrate
BUD Bottom and up-and-down stream sides open	3	Movement vertically downward and upward and up-and-downstream movement within the substrate
SUD Sides and up-and-downstream sides open	4	Lateral movement and up-and-downstream movement within the substrate
C All sides open (control)	6	Movement from all directions

tions: (1) horizontal up-and-downstream within-substrate movement (parallel to stream flow), (2) horizontal lateral movement (perpendicular to stream flow), (3) vertical movement (toward and away from the hyporheic zone) and (4) drift (movement into the substrate from the overlying water column). For each experiment, 50 substrate-filled cages (10 each of 5 types) were buried at each riffle. Cages were made from 1 pint rectangular plastic containers with dimensions $9.5 \times 9.5 \times 8.5$ cm and had removable plastic lids. Openings, 6.5×6.5 cm, were cut in two to six sides to allow colonization of invertebrates and covered with 1.25 cm mesh hardware cloth to retain substrate. Stones collected from the streambed and ranging in size from 1.5 cm to 10 cm largest diameter, were placed in the cages. The proportion of stones of each size placed in the cages was determined by measuring all the stones to a depth of 10 cm in a 0.3 m^2 area of streambed. All cages were open in the up-and-downstream directions so that flow conditions would be nearly uniform between cages.

Cages were buried in the stream with the tops flush with the surrounding streambed at sites randomly chosen from a grid pattern with 300 available places (3 across and 100 lengthwise over a 15 m section). However, cages were never placed directly next to or upstream from another cage.

Cages were left in the stream for 3 days, long enough to detect differences in directional movements, but short enough to prevent total colonization (3 days minimum, Delucchi 1988a), which might mask directional movements, and to minimize cage effects (Peckarsky and Penton, personal communication). Turbidity caused by a thunderstorm during the removal of cages at the temporary stream riffle prevented the removal of 15 cages at that site. Depth and current were measured at each cage when it was buried and again when it was removed (Table 2). An aquarium net (maximum mesh opening $480 \mu\text{m}$) was held directly downstream of cages and then placed under the cages while they were removed from the stream and placed in a pan

of water; the cage, stones and large detritus were rinsed and removed. The remaining detritus and invertebrates were strained through the aquarium net and preserved in 70% ethanol. Samples were sorted under a dissecting microscope at $10 \times$ magnification.

To compare the density of invertebrates colonizing cages to that of the surrounding benthos, five Surber samples ($355 \mu\text{m}$ mesh) were taken on the last day of the experiment at each riffle at sites selected randomly in the grid pattern. Surber samples were processed in the same manner as cage samples in the field and laboratory except that those containing over 300 invertebrates were subsampled using a method modified from Wrona et al. (1982) with 1/4 of the sample counted and identified.

Migration experiment

In September 1986, experiments were run at three riffle sites (one in the permanent stream, and a permanent and a temporary riffle in the intermittent stream (Table 1)) during the 10 days immediately prior to the drying of the temporary riffle. Three cages (one allowing upstream movement, one downstream movement and a control cage allowing flow but restricting invertebrate colonization) were buried at each end of each riffle to monitor invertebrate movement into and out of the riffle. Cages were made from plastic freezer containers with the dimensions 24 (length) $\times$ 13 (width) $\times$ 10.5 cm (height). One opening (10×7 cm) was cut on up-and-downstream ends and two on each of the longer sides. Openings were covered with mesh ($260 \mu\text{m}$) on three of four sides for directional movement cages and on all four sides for control cages. These openings enabled current to flow through cages while limiting invertebrate colonization from non-target directions. The remaining opening on movement cages was covered with 1.5 cm mesh hardware cloth to retain substrate but allow colonization.

Cages were buried in random order across the ends of each riffle. One drift net (mouth 10.5×10.5 cm, $440 \mu\text{m}$ mesh) was placed on a randomly selected cage. Cages and drift nets were left in place for approximately 24 h, at which time they were removed and processed as were cages for the previous experiments. Cages were reburied immediately after processing. This was repeated for 8 of the 10 days prior to the drying of the temporary riffle. Downstream movement data were not available for the eighth day at the permanent stream riffle and the temporary riffle because of an error in cage orientation. Current and depth were measured at each cage at burial and retrieval (Table 4). Each day, discharge and dissolved oxygen were measured at each end of each riffle (Fig. 1, Table 4).

To compare within-cage flows between cage types, hemispheres of plaster of Paris were placed in each cage at burial. The hemispheres were dried and weighed before and after placement in cages. Since dissolution of hemispheres has a linear relationship with current (Peckarsky and Penton, personal communication), the difference in weights was used as a relative measure of current in cages. A Kruskal-Wallis test showed a significant difference between cage types ($N=129$, $P<0.05$). However, a Mann-Whitney test showed that upstream and downstream movement cages were not different ($N=86$, $P>0.05$).

Invertebrates collected in cages were classified as large ($>360 \mu\text{m}$ maximum width) or small ($<360 \mu\text{m}$ maximum width) according to whether or not they passed through

Table 4. Ranges of abiotic conditions measured at each riffle end for the migration experiment (September 1986). Definitions of riffle and stream types are given in Table 1

Stream type Riffle type	Permanent Permanent		Intermittent Permanent		Intermittent Temporary	
Riffle end	Upstream	Downstream	Upstream	Downstream	Upstream	Downstream
Range of current (m/s)	0.17–0.33	0.08–0.47	0.02–0.12	0.02–0.24	0.02–0.14	0–0.24
Range of depth (cm)	5.5–10.5	5.5–9.5	4.0–8.0	3.0–6.0	1.5–4.0	0–4.0
Range of O ₂ (% saturation)	93–103	91–103	93–97	97–101	92–99	90–99

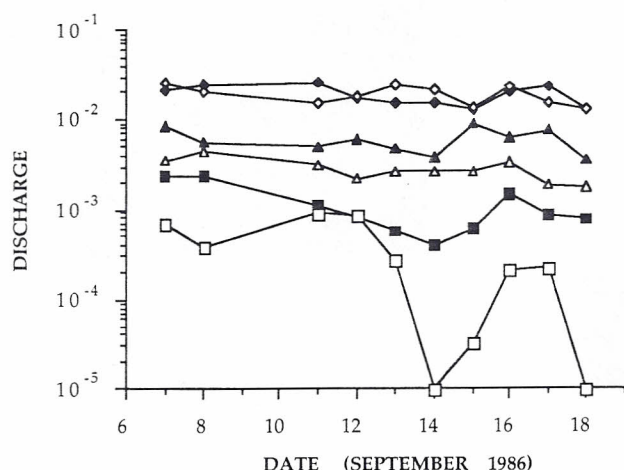


Fig. 1. Discharge (m^3/s) at upstream and downstream ends of the three riffles used for the migration experiment. Definitions of riffle and stream types are in Table 1. Permanent stream $\blacklozenge$ upstream $\diamond$ downstream, intermittent stream permanent riffle $\blacktriangle$ upstream $\triangle$ downstream, temporary riffle $\blacksquare$ upstream $\square$ downstream

the diagonal of the small mesh on cages. Large and small invertebrates were analyzed separately since small invertebrates could colonize from non-target directions and were, therefore, disproportionately represented in control cages. Before data analysis correction factors (numbers of invertebrates in each size and taxon in the adjacent control cage) were subtracted from actual numbers obtained in movement cages.

Results

Within-riffle movement experiment

For each riffle a planned linear contrast was used to compare the control cages to movement cages to test for additiveness of colonization from different directions. In all riffles, up-and-downstream movement (UD)+drift (TUD-UD)+vertical movement (BUD-UD)+lateral movement (SUD-UD) was not significantly different from colonization of control cages indicating that colonization from different directions was additive (t-test, $T=0.17$ – 1.35 , $N=35$ – 50 , $P>0.5$ – 0.1).

A Kruskal-Wallis test indicated that at four of the five riffles there was a significant difference in the number of invertebrates colonizing the different cage types (Fig. 2). There was no difference in colonization among cage types at the dry riffle. Since all cages were open in up-and-down-

stream directions, this result indicates that movement from all other directions was swamped by up-and-downstream movement. The streambed immediately downstream from this study riffle was dry during the experiment. The number of invertebrates colonizing cages was negatively correlated with depth in this riffle ($R=-0.44$, $P<0.01$, $N=49$), which decreased towards the downstream end of the riffle (the dry area). These results suggest that invertebrate movement was highest in the shallow area closest to the drying section of the riffle. This experiment does not, however, enable a distinction between up-and-downstream movement.

For the riffles that did show significant differences among colonization of cage types, three planned linear contrast were made to test for significant movement in directions other than up-and-downstream. The contrasts made used Mann-Whitney tests to determine differences between up-and-downstream movement cages and each of the other three types of movement cages (TUD, BUD, and SUD) (Fig. 2). At the riffle on the permanent stream, drift cages (TUD) had significantly more invertebrates than up-and-downstream movement cages (UD) ($N=19$, $P=0.0022$). At the permanent riffle on the intermittent stream, drift and lateral movement (SUD) cages had significantly more ($N=20$, $P=0.0006$) and $N=20$, $P=0.0173$, respectively) and vertical movement (BUD) cages significantly fewer ($N=20$, $P=0.0015$) invertebrates than did up-and-downstream cages. At the temporary stream riffle, vertical movement cages also had significantly fewer invertebrates than up-and-downstream cages ($N=14$, $P=0.041$). At the temporary riffle on the intermittent stream, no movement cages had a significantly different number of invertebrates than the up-and-downstream cages ($N=20$, $P>0.17$); however, drift cages had more invertebrates than vertical movement cages (unplanned linear contrast, Mann-Whitney test, $N=20$, $P=0.0025$). In summary, movement vertically downward toward the hyporheic zone was important at the permanent riffle on the intermittent stream and the temporary stream riffle, while drift was important only at permanent riffles. However, at riffles that had decreasing flow during the experiments, the temporary and dry riffles on the intermittent stream, up-and-downstream movement dominated.

Taxa that occurred in densities of greater than one per cage were analyzed in the same manner as total invertebrates. Individual taxa that showed significant movement in directions other than up-and-downstream generally showed the same pattern as that of the total invertebrates, with drift being important at the permanent riffles and vertically downward movement being important at both permanent and temporary riffles (Table 5). Although total invertebrates showed no movement vertically downward at the temporary stream riffle, *Paraletophlebia* (Leptophlebiidae:

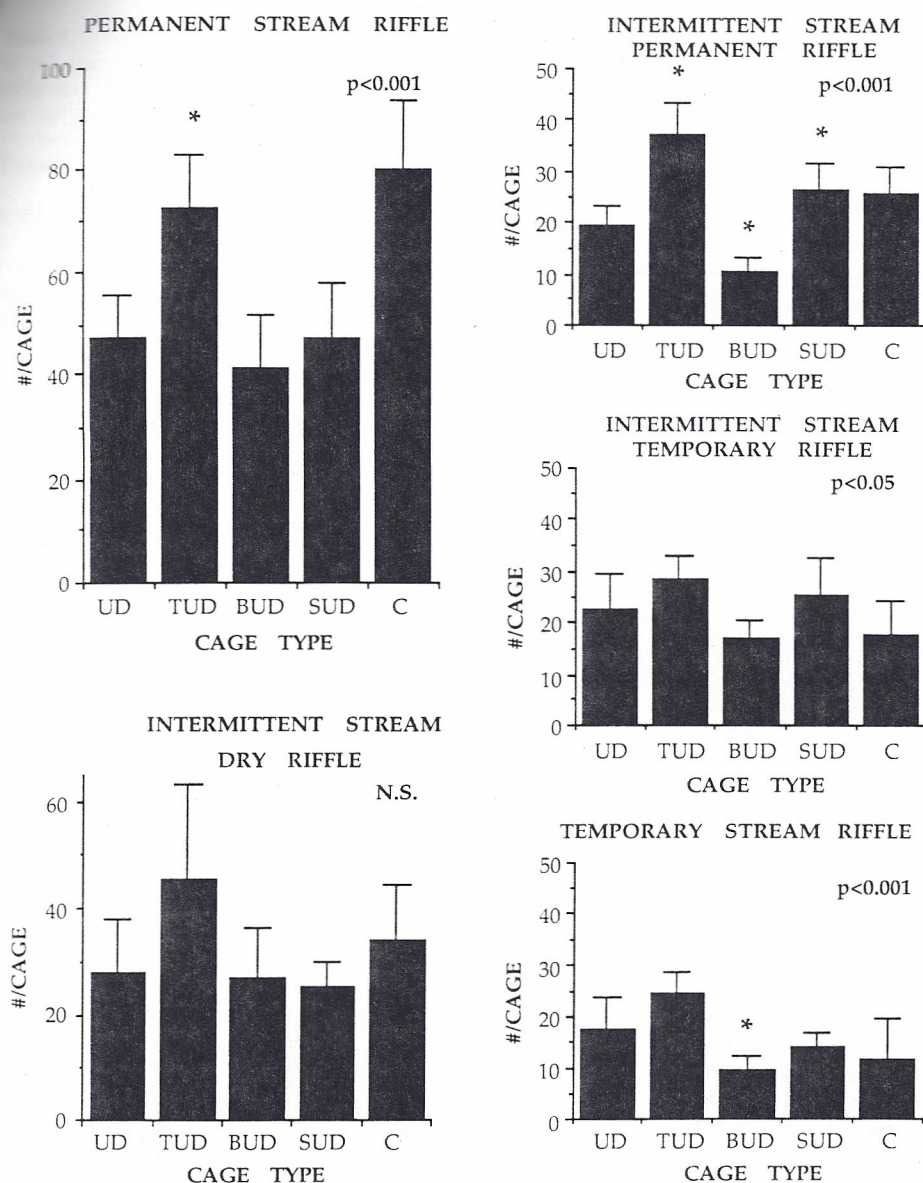


Fig. 2. Average number of invertebrates per cage for each of the five cage types at the five different riffles in the within-riffle movement experiment. *P*-values for significant Kruskal-Wallis tests are given. N.S. = not significant. Asterisks indicate movement cage types with a significantly different (Mann-Whitney test, $P < 0.05$) number of invertebrates than up-and-downstream movement cages. Cage type abbreviations as in Table 3. Error bars are ± 2 SE

Ephemeroptera) and Chloroperlidae (Plecoptera), did show such movement. Chironomids (Diptera) had different movement patterns at different riffles, which may be due to differences in species composition between riffles rather than to behavioral differences. *Paraleptophlebia* spp. (mainly *P. mollis*) showed consistent behavior between riffles, having a tendency to move vertically downward at all riffles. Surprisingly, the *Baetis* (Baetidae: Ephemeroptera) species were not responsible for the large amount of drift at the permanent stream riffle even though they were most abundant at that riffle and usually comprise a high proportion of drifting stream invertebrates (Wiley and Kohler 1984).

Chi-square analysis was used to compare the proportion of abundant taxa in cages to their abundance in Surber samples. Approximately half the taxa tested were overrepresented and half were underrepresented in cages. Taxa were consistent in their patterns from site to site; chironomids and *Paraleptophlebia* showed generally high rates of movement, while baetids and *Leuctra* (Leuctridae: Plecoptera) showed low rates of movement. One exception was Oligo-

chaetes, which were underrepresented in cages at the permanent riffle on the intermittent stream yet overrepresented in cages at the temporary and dry riffles on the same stream. Thus, Oligochaetes showed higher rates of movement at riffles that dry.

Even though cages were left in the streams for only 3 days, densities in cages at riffles on the intermittent and temporary streams had reached or exceeded ambient levels (as determined by Surber samples) at the end of 3 days (Fig. 3). This was not true in the permanent stream even though the absolute number of invertebrates colonizing cages was highest in this stream. Thus, invertebrates in streams susceptible to drying seemed to freely colonize available space faster than those in the permanent stream. These results are consistent with those of a previous study (Delucchi 1988a) that showed higher absolute colonization rates (number of individuals in cages) in the permanent stream in mid-summer but higher relative colonization rates (number colonizing relative to the density of the surrounding benthos) in the intermittent stream during the same time period.

Table 5. Results of Kruskal-Wallis test for differences in number of colonizers in different cage types for taxa that occurred in densities of >1 per cage. Abbreviations under P value for Kruskal-Wallis test indicate the movement cage types (Table 3) that were significantly different (Mann Whitney test, $P < 0.05$) from up-and-downstream movement cages and direction of difference. Numbers in parentheses are the sample sizes for Mann-Whitney tests

Stream type	Permanent	Intermittent			Temporary
Riffle type	Permanent	Permanent	Temporary	Dry	Temporary
N	49	50	50	49	35
<i>Baetis tricaudatus</i>	N.S.				
<i>Baetis flavistriga</i>	N.S.				
<i>Baetis hageni</i>	N.S.		*** TUD(+)(20)		
<i>Epeorus</i>					N.S.
<i>Paraleptophlebia</i>		** BUD(-)(20)	N.S.		*** BUD(-)(14)
Leptophlebiidae	N.S.				
<i>Ameletus ludens</i>				N.S.	N.S.
<i>Tricorythodes</i>	N.S.				
Chloroperlidae			*		*** BUD(-)(14)
<i>Leuctra</i>					N.S.
Hydropsychidae	N.S.				
Chironomidae ^a	*** TUD(+)(19)	*** TUD(+)(20) BUD(-)(20)	N.S.	N.S.	** SUD(-)(17)
Tanypodinae	*** TUD(+)(19)	*** BUD(-)(20)	N.S.	*	
<i>Prosimulium</i>	N.S.				
Diptera pupa	N.S.	N.S.			
Oligochaeta			*	N.S.	
<i>Torrenticola</i>	* TUD(+)(19)				

^a Includes Orthocladinae and Chironominae; Tanypodinae analyzed separately

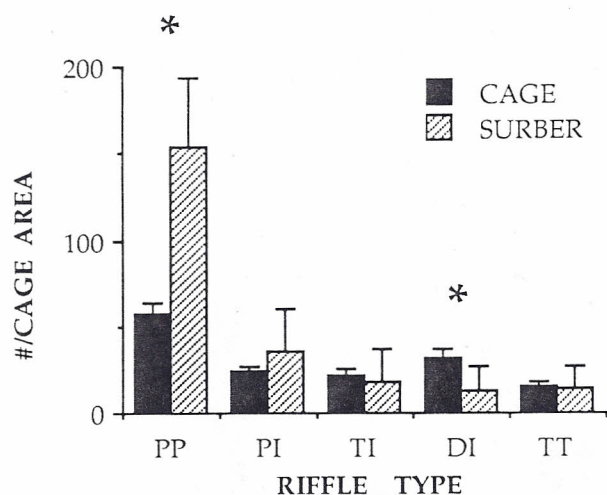


Fig. 3. Average number of invertebrates in cages and per cage area (90 cm^2) in Surber samples at each riffle for within-riffle movement experiment. PP permanent stream riffle, PI permanent riffle on the intermittent stream, TI temporary riffle on the intermittent stream, DI dry riffle on the intermittent stream, TT temporary stream riffle. Asterisks indicate pairs that are significantly different (Mann Whitney test, $P < 0.05$). Error bars are $\pm 2 \text{ SE}$

Migration experiment

While the previous experiment examined generalized movement patterns throughout a riffle, the migration experiment was designed to distinguish upstream from downstream

movement and measure movement into and out of riffles during drying. During the course of the experiments discharge decreased significantly (rank correlations, $N=10$, $P < 0.05$) at the downstream end of both riffles on the intermittent stream, but not on the permanent stream (Fig. 1). The experiment was ended when the downstream end of the temporary riffle dried completely.

Drift rate ($\text{no}/100 \text{ m}^3$) did not differ between up-and-downstream ends of any riffle. Total number of invertebrates entering and leaving riffles through drift ($\text{no}/24 \text{ h}$) was also not significantly different at the permanent riffles; however, there was significant net movement into the temporary riffle (Wilcoxon signed rank test, $N=8$, $P=0.008$) (Fig. 4A). This was a result of there being higher discharge into than out of the riffle (Fig. 1) while drift densities remained constant.

Within-substrate movement into and out of riffles also did not differ at the permanent riffles. However, there were significantly more large invertebrates moving upstream at the downstream end of the temporary riffle as compared to the upstream end of the riffle (Wilcoxon signed rank test, $N=8$, $P=0.022$) (Fig. 4B). *Paracapnia opis* (Capniidae: Plecoptera), *Leuctra* spp. and Tanypodinae (Chironomidae: Diptera) were the major contributors to this pattern. Thus, at the temporary riffle both drift and upstream movement contributed to net movement into the drying riffle.

Comparisons of the general directional movement pattern between streams show that upstream movement and downstream movement within the substrate were of equal

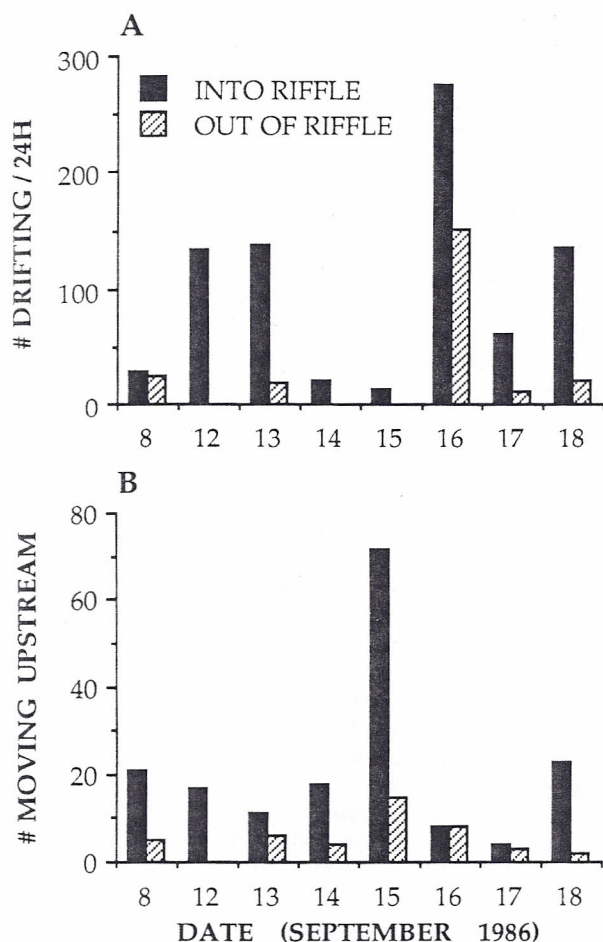


Fig. 4A, B. Number of (A) total invertebrates drifting and (B) large ($> 360 \mu\text{m}$ maximum width) invertebrates moving upstream into and out of the temporary riffle on the intermittent stream during the migration experiment

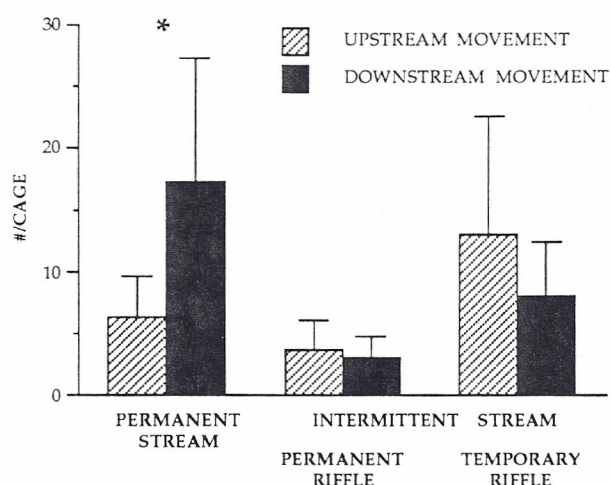


Fig. 5. Average number of invertebrates moving upstream and downstream at three different riffle types during the migration experiment. Asterisk indicates a significant difference between up- and downstream movement (Wilcoxon signed rank test, $P=0.006$)

magnitude in the intermittent stream (Wilcoxon signed rank test: $N=16$, $P=0.820$, $N=15$, $P=0.081$, permanent and temporary riffles, respectively). In the permanent stream downstream movement was of much higher magnitude than

upstream movement (Fig. 5) (Wilcoxon signed rank test, $N=15$, $P=0.006$). In addition, drift rates were much lower (Mann-Whitney test, $P<0.001$) in the intermittent stream (median = $38/100 \text{ m}^3$) than the permanent stream (median = $261/100 \text{ m}^3$).

Discussion

Movement patterns in the stream that is permanent along its entire length, conformed to what has been found in other studies (Bishop and Hynes 1969; Elliot 1971; Peckarsky 1979). Downstream movement, both within-substrate and by drift, was the dominant movement pattern. In addition, movement was essentially unidirectional, as lateral movement and vertical movement toward or away from the hyporheic zone were insignificant and upstream movement was much lower than downstream movement. However, there was no net loss of invertebrates from riffles; riffles gained as many invertebrates from upstream reaches as they lost through downstream movement.

Movement in the streams that have riffles susceptible to drying showed a different pattern than that of the permanent stream. The proportion of the benthos colonizing cages was higher relative to that in the permanent stream as shown by the comparison of invertebrate densities in cages in the within-riffle movement experiment with Surber sample densities (Fig. 3). This result is consistent with that of a previous study (Delucchi 1988a) for this time of year (beginning of drying); although, the relationship changed during maximum dryness and reflooding. During these later periods both absolute and relative number of invertebrates colonizing cages were higher in the permanent stream. Further, movement vertically downward toward the hyporheic zone and upstream movement substantially contributed to a departure from the unidirectional movement pattern seen in the permanent stream. Interstitial flow rates may influence the amount of movement into the hyporheic zone (Williams 1984). Since stream invertebrate movements are generally responsive to current (Ciborowski and Clifford 1984; Townsend and Hildrew 1976; Ulfstrand 1968), a high flow rate vertically downward between the stream and the hyporheic zone might lead to increased movement of invertebrates. Figure 1 provides evidence that the intermittent stream might have this type of flow pattern. In this stream, discharge into riffles was always higher than discharge out of riffles. This observation is consistent with the hypothesis that water was lost to the hyporheic zone. A further observation that riffles always dry starting from the downstream end also provides evidence to support this hypothesis.

This drying pattern has implications for the usefulness of drift as a mechanism by which invertebrates can escape drying. Individuals that drift while the stream is drying will more probably be concentrated in a drying riffle. Interestingly, the riffle in the intermittent stream that is dry for the longest period of time (dry riffle) is downstream of all other temporary and permanent riffles, preventing movement from the intermittent to permanent stream for over 3 months of the year. Thus, the drift pattern observed in this study (low rates in the intermittent stream) would be an advantage in this habitat.

There are two mechanisms potentially responsible for low drift rates in the intermittent stream. First, since there is generally a positive relationship between invertebrate drift rate and current or discharge (Waters 1972) the low

drift rate in the intermittent stream may be a passive response to low discharge in this stream (Fig. 1). However, two previous drift studies comparing intermittent and permanent streams (Delucchi 1988a; Dance and Hynes 1979) have shown that while there is a positive relationship between drift and discharge in permanent streams, there is no relationship between discharge and drift in intermittent streams. Alternatively, the lower drift rates in the intermittent stream could have been due in part to differences in species composition between streams. Of the three species of *Baetis* (a genus that makes up a large proportion of the drift in these and other streams (Waters 1972)) that occur in the stream, the two that are found in higher proportion in the drift relative to benthic densities (*B. tricaudatus* and *B. flavistriga*) (Delucchi 1988a), i.e., have a higher propensity to drift, are also found in lower densities in the benthos of the intermittent stream than the third species (*B. hageni*), which has a low propensity to drift (Delucchi 1988a). Thus, the drift pattern observed in the intermittent stream is due in part to the dominance of species with a lower propensity to drift, regardless of the habitat type.

The rate of within-substrate invertebrate movement also showed a different pattern in streams susceptible to drying than in the permanent stream. The migration experiment, which was carried out as the streams were actually drying, showed the highest rate of invertebrate movement in the specific area that was drying. This result is consistent with the higher rate of colonization at the downstream (drying) end of the dry riffle during the within-riffle movement experiment. However, other evidence suggests that there is not a simple relationship between decreasing current and increased movement. For example, the downstream end of the permanent riffle on the intermittent stream had decreasing discharge during the migration experiments, but no difference in movement between upstream and downstream ends of the riffle. In addition, previous colonization studies at the permanent stream site and the permanent site on the intermittent stream showed that the lowest rates of movement occurred during the time of maximum decrease in discharge (Delucchi 1988a). Thus, increased movement during decreased discharge only occurred at sites where drying was imminent.

Even though invertebrates showed increased movement during drying, movement was in a direction that would result in their concentration in the drying riffle unless they subsequently moved into the hyporheic zone. However, the present study provides only weak evidence of hyporheic movement in temporary riffles. In addition, riffles that had the largest decreases in current and discharge during the experiment and, therefore, were most probably closest to drying (the dry and temporary riffles on the intermittent stream) did not show significant vertical movement of invertebrates, suggesting that invertebrates are not moving into the hyporheic zone during drying. Presumably the individuals remaining when streams dry are population variants or species that will not survive drying. On the other hand, previous studies conducted in these riffles provide evidence that other individuals survived by leaving riffles before drying was imminent. For example, although the most dramatic difference in community structure between the permanent and intermittent streams is between June samples and those taken after drying and reflooding (September), a change does occur between June and July (Delucchi 1988b). Samples taken in July were intermediate in community struc-

ture, indicating that communities were in a transitional stage, with initial changes in composition occurring prior to drying. This change is due primarily to emergence before drying of several of the species abundant in the intermittent stream (Delucchi 1988a).

However, both this and previous studies (Delucchi 1988a, b) in these streams show that taxa with no specific features ensuring survival in temporary streams do occur in these streams and most probably become stranded and die during drying. This result is in contrast to Williams and Hynes (1977) observation in temporary streams in Ontario that there is "little wastage of species or individuals" even when temporary and permanent streams are connected. I would argue that, in many temporary streams there may not be great pressure for inhabitants to avoid drying for two reasons. (1) Periodic drying events will never entirely eliminate them from a stream because individuals that are in permanent refuges, such as, non-drying riffles and pools, the hyporheic zone and permanent streams, may provide sufficient numbers of individuals to repopulate the stream after reflooding. (2) The extent of drying in a temporary stream can vary greatly from year to year (Delucchi 1988a); in wet years, species that do not leave the stream before the normal dry period may be able to utilize the extended season to reproduce in the stream, thereby attaining higher benthic densities in the temporary stream during the following year. Thus, although populations might be expected to fluctuate between years depending on precipitation, high migration and reproductive rates would allow species without specific drought resistant characteristics to maintain populations in temporary streams.

In summary, this study shows that movement patterns of invertebrates in permanent and temporary streams differed. Invertebrate movement in the permanent stream was dominated by drift and downstream within-substrate movement. In contrast, invertebrates in streams susceptible to drying showed additional movement in the upstream direction and vertically downward toward the hyporheic zone. Increased movement into the hyporheic zone may be facilitated by a large exchange of water between streambed and hyporheic zone, a pattern of water flow that may also increase the susceptibility of the stream to drying. Drift, a potential mechanism by which invertebrates can migrate from intermittent riffles, was very low in the intermittent stream and was only a significant factor at permanent riffles. The fact that riffles dry from the downstream end up reduces the probability that drift is an important mode of transportation for invertebrates out of drying riffles. Up-and-downstream movement, especially upstream movement, dominated as the stream was actually drying but not at the end of the riffle where it would lead to migration into non-drying refuges.

In conclusion, even though this study provides some evidence that movement patterns potentially resulting in migration by invertebrates out of drying riffles exist, e.g., movement into the hyporheic zone, the movement patterns observed during drying would result in concentration and subsequent death of invertebrates remaining in drying riffles. Thus, the patterns of invertebrate movement in temporary and permanent riffles did not support the hypothesis that a fauna exists with behaviors specific to survival in temporary streams. Movement patterns observed are most probably the result of invertebrate responses to differences in hydrology between streams.

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